

Lab Assignment 1: Environment Setup, Preprocessing and Exploratory Data Analysis

CSAI2017P: Applied Machine Learning (Lab)

Experiments 1–2 (Units I, IV) • CO1, CO2

Instructor: Dr. Tofik Ali

Due: after lab session 2

Student Name: _____

SAP ID: _____

Batch: _____

Lab quiz: LQ1, lab session 3

Objective

- Stand up a reproducible Python ML environment and prove it works.
- Load a tabular dataset and describe it before modelling anything.
- Diagnose missing values, outliers and scale problems, and treat them.

Datasets used in this assignment

A1 (California Housing), A2 (Telco Customer Churn).

Full descriptions, sources and loading snippets for all anchor datasets are in **Anchor-Datasets.pdf**, alongside this brief.

This assignment runs ahead of the lectures

Experiment 2 draws on Unit IV (data preprocessing), which is lectured around sessions 9–14. You are not expected to know it yet. Everything you need is in the Lecture 1 notes (§10, the Python ML stack) and the scikit-learn preprocessing guide linked there. Ask in the lab rather than guessing.

Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

Coverage rule for this assignment. Both datasets must appear in your submission. If you skip A3 in Section A, you must answer B2 in Section B.

Submit:

- Notebook: `AML_LA01_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.

- Do not upload datasets. Cite the source and include the loading code.

Section	Level	Choice rule	Marks
A	Easy	Attempt any 2 of 3 (2 marks each)	4
B	Medium	Attempt any 1 of 2 (4 marks each)	4
C	Consolidation	Compulsory	2
Total			10

Course outcomes assessed by this assignment: **CO1, CO2**. Attempting more than the choice rule allows is not penalised; the best attempts are marked. Component weightings are indicative and subject to departmental approval; the confirmed split will be posted on the LMS.

Section A (Easy) — attempt any 2 of 3 ($2 \times 2 = 4$)

A1. Environment proof [2 marks]

No dataset.

- Create a fresh environment and install `numpy`, `pandas`, `scikit-learn`, `matplotlib`, `seaborn`, `jupyter`.
- Print the version of each of the five libraries and your Python version in one cell.
- State in two lines why pinning versions matters when you hand work to someone else.

A2. First look at the data [2 marks]

Dataset: A1 (California Housing).

- Load with `fetch_california_housing(as_frame=True)` and run `head`, `info`, `describe` and `isna().sum()` on `d.frame`.
- State the number of rows, the number of features, the target, and its units. (`print(d.DESCR)` tells you the units — the `return_X_y` form throws them away.)
- Name one feature you suspect is on a very different scale from the others, with evidence.

A3. Missing values and outliers [2 marks]

Dataset: A2 (Telco Customer Churn).

- Find the column that looks numeric but is stored as text, and explain why.
- Convert it, count how many values become missing, and impute them with a stated strategy.
- Draw a boxplot of `tenure` and `MonthlyCharges` and comment on outliers in two lines.

Section B (Medium) — attempt any 1 of 2 ($1 \times 4 = 4$)

B1. Scaling changes the answer [4 marks]

Dataset: A1 (California Housing).

- Split 80/20 with `random_state=0`. Train a `KNeighborsRegressor(n_neighbors=5)` on the raw features and report test MAE.

- (b) Repeat inside a `Pipeline` with `StandardScaler`. Report test MAE again.
- (c) Explain in 4–6 lines why the difference appears, referring to how KNN measures distance.

B2. Correlation and redundant features [4 marks]

Dataset: A2 (Telco Customer Churn).

- (a) Convert `TotalCharges` with `pd.to_numeric(df['TotalCharges'], errors='coerce')`, then encode `Churn` as 0/1.
- (b) Compute `df.corr(numeric_only=True)` — the plain `df.corr()` raises `ValueError` on this frame because of the text columns. Rank the four numeric features against `Churn` and report the strongest three.
- (c) `TotalCharges` is roughly `tenure×MonthlyCharges`. Explain in 4–6 lines what problem that redundancy causes, and what you would do about it.

Section C (Consolidation) — compulsory (2)**C1. Reproducibility check** [2 marks]

Either dataset.

- (a) Restart the kernel and run your notebook top to bottom.
- (b) State whether every number in your report reproduced exactly. If not, identify which and why.
- (c) Now change `random_state` from 0 to 1 and re-run one model. Report how much your headline metric moved, and say in two lines what that tells you about how precisely you should quote it.

End of Lab Assignment 1

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.; McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.; Scikit-learn user guide, accessed 2026-08-04.; Matplotlib and Seaborn documentation, accessed 2026-08-04.